

SKIN MODULE

Quick Review

Use this sheet to review the big ideas from *Meet Your Skin* and *How Skin Heals*.

1. Read the skin from the outside in

EPIDERMIS	DERMIS	HYPODERMIS
The avascular outer barrier. Keratinocytes make keratin and continually renew its surface.	The vascular support layer. It contains blood vessels, nerves, collagen, follicles and glands.	The subcutaneous layer beneath the skin. Fat and connective tissue cushion, insulate and anchor.

2. Follow the overlapping repair stages

1 STOP THE LEAK	2 CLEAN AND DEFEND
Blood vessels narrow, platelets gather and a clot forms a temporary seal.	Immune cells remove damaged material and coordinate inflammation.
3 REBUILD	4 STRENGTHEN
New surface cells, blood vessels, collagen and granulation tissue restore the wound.	Collagen reorganizes for weeks or months. A scar may flatten and fade.

Key idea: healing stages overlap. Depth, location, health, infection and care all affect timing.

3. Protect the repair zone

CARE FOR A MINOR CUT OR SCRAPE	GET MEDICAL HELP
- Wash your hands. - Use gentle pressure to stop bleeding. - Rinse with cool running water. - Cover with a clean bandage. - Watch whether it improves.	Seek help for bleeding that will not stop, deep or gaping wounds, embedded objects, animal or human bites, or infection signs such as spreading redness, worsening pain, pus, fever or red streaks. Dirty or serious wounds may need a tetanus assessment.

CHECK YOUR UNDERSTANDING

1. Why can a dermal injury bleed, but a shallow epidermal injury may not?
2. Put the four healing stages in order. Why is that order only a model?
3. Name two jobs of the hypodermis.
4. What do fibroblasts produce during repair?
5. Which changes suggest that a wound is worsening instead of healing?

IMPORTANT DEFINITIONS

Alphabetical Glossary

Scientific terms from the anatomy and wound-healing lessons.

Adipose tissue - Connective tissue made mostly of fat cells. It stores energy, insulates and cushions.

Apocrine gland - A sweat gland that empties into a hair follicle in selected areas and becomes active at puberty.

Capillary - The smallest blood vessel. It exchanges oxygen, nutrients, heat and wastes with tissue.

Clot - A temporary plug made from platelets and a fibrin mesh that helps stop bleeding.

Collagen - A strong structural protein that supports connective tissue and forms much of a scar.

Connective tissue - Supporting tissue that binds, cushions and strengthens body structures.

Cutaneous receptors - Nerve endings and specialized sensors that detect touch, pressure, pain and temperature.

Dermis - The vascular connective-tissue layer beneath the epidermis. It contains nerves, vessels, follicles and glands.

Ecocrine gland - A widespread sweat gland that opens onto the skin surface and helps cool the body.

Elastin - A flexible protein that helps skin stretch and return toward its original shape.

Epidermis - The avascular, keratinized outer layer of the skin.

Epithelialization - The movement and growth of new epithelial cells across a wound surface.

Exocrine gland - A gland that releases its product through a duct to a surface or cavity.

Fascia - Sheets of connective tissue that surround and connect muscles and other structures.

Fibroblast - A repair cell that produces collagen and other extracellular-matrix materials.

Granulation tissue - New repair tissue containing small blood vessels, fibroblasts and extracellular matrix.

Hair follicle - A tube-like pocket in the skin from which a hair grows.

Hemostasis - The body's process for stopping bleeding through vessel narrowing, platelets and clotting.

Hypodermis - The subcutaneous tissue beneath the skin that cushions, insulates and anchors it.

Inflammation - A coordinated immune response that removes damaged material and signals repair.

Integumentary system - Skin plus its supporting structures, including hair, nails and glands.

Keratin - A tough structural protein found in the epidermis, hair and nails.

Keratinocyte - The main epidermal cell. It makes keratin and helps build the skin barrier.

Melanin - A pigment that contributes to skin colour and helps protect cellular DNA from ultraviolet radiation.

Melanocyte - A pigment-producing epidermal cell that makes melanin.

Nail matrix - The growth zone that produces the hard keratin cells of the nail plate.

Platelet - A small blood component that sticks at an injury and helps begin clot formation.

Remodeling - The long strengthening stage in which collagen is reorganized and a scar changes.

Sebaceous gland - An oil gland that releases sebum, usually into a hair follicle.

Sebum - The oily product that lubricates skin and hair and supports the surface barrier.

Scar - Fibrous repair tissue that replaces normal tissue after a deeper injury.

Stratum basale - The deepest epidermal layer, where dividing cells replace keratinocytes.

Stratum corneum - The outermost epidermal layer, made of flattened keratin-rich cells within a lipid barrier.

Subcutaneous - Located under the skin; another term used for the hypodermal tissue level.

Sudoriferous gland - The scientific name for a sweat gland.

Review date: September 2026. Based on the Homeschoolable Meet Your Skin and How Skin Heals modules.